Experiment 2: Strassen Matrix Multiplication Algorithm

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #2 | Academic Year: 2026-27

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1. Overview & Objective:

Divide-and-conquer implementation reducing 8 submatrix multiplications to 7 for O n^2.807 execution.

2. Practical Tasks & Verification Steps:

1. Implement standard matrix multiplication with three nested loops in O n^3 time.
2. Partition nˆ n square matrices into four n/2 ˆ n/2 submatrices.
3. Compute 7 Strassen products P1 through P7 using submatrix additions and subtractions.
4. Combine products into final result matrix and compare CPU operations against standard multiplication.

Submission Guidelines & Verification:

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